

Ex-1

```
Ex-1.cpp > main()
4  int main()
5  {
6      ofstream dataFile;
7      dataFile.open("data.txt");
8      string name;
9      int age;
10     if (!dataFile)
11     {
12         cout << "Error in Opening file" << endl;
13     }
14     else
15     {
16         cout << "Enter Your Name: ";
17         getline(cin, name);
18         cout << "Enter Your Age: ";
19         cin >> age;
20         dataFile << "Name: " << name << " Age: " << age << endl;
21         dataFile.close();
22         cout << "Data Written to File Successfully" << endl;
23     }
24 }
```

```
Enter Your Name: Yash
Enter Your Age: 16
Data Written to File Successfully
```

FILE:

```
data.txt
1  Name: Yash Age: 16
2  |
```

TASK-1

```
Task-1.cpp > main()
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4  int main()
5  {
6      ofstream dataFile;
7      dataFile.open("data.txt", ios::app);
8      string name;
9      int age;
10     if (!dataFile)
11     {
12         cout << "Error in Opening file" << endl;
13     }
14     else
15     {
16         cout << "Enter Your Name: ";
17         getline(cin, name);
18         cout << "Enter Your Age: ";
19         cin >> age;
20         dataFile << "Name: " << name << " Age: " << age << endl;
21         dataFile.close();
22         cout << "Data Written to File Successfully" << endl;
23     }
24 }
```

```
Enter Your Name: Khunais Baig
Enter Your Age: 25
Data Written to File Successfully
```

FILE:

```
data.txt
1  Name: Yash Age: 16
2  Name: Khunais Baig Age: 25
3
```

Ex-2

```
Ex-2.cpp > main()
1  #include <iostream>
2  #include <fstream>
3  #include <string>
4  using namespace std;
5
6  int main()
7  {
8      ofstream numFile;
9      numFile.open("numbers.txt", ios::app);
10     if (!numFile)
11     {
12
13         cout << "Error in opening file!" << endl;
14         return 1;
15     }
16     else
17     {
18         int num;
19         for (int i = 0; i < 10; i++)
20         {
21             cout << "Enter Number " << i + 1 << " : ";
22             cin >> num;
23             numFile << num << endl;
24         }
25         numFile.close();
26         cout << "Numbers Written to File Successfully" << endl;
27     }
28 }
```

```
29     ifstream readFile;
30     readFile.open("numbers.txt");
31     if (!readFile)
32     {
33         cout << "Error in opening file!" << endl;
34         return 1;
35     }
36     else
37     {
38         int number;
39         cout << "EvenNumbers from File:" << endl;
40         while (readFile >> number)
41         {
42             if (number % 2 == 0)
43             {
44                 cout << number << endl;
45             }
46         }
47         readFile.close();
48     }
49 }
50
```

OUTPUT:

```
Enter Number 1 : 2
Enter Number 2 : 6
Enter Number 3 : 8
Enter Number 4 : 10
Enter Number 5 : 12
Enter Number 6 : 14
Enter Number 7 : 16
Enter Number 8 : 17
Enter Number 9 : 20
Enter Number 10 : 22
Numbers Written to File Successfully
EvenNumbers from File:
2
6
8
10
12
14
16
20
22
```

TASK-2

```
Task-2.cpp > main()
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4
5  int main()
6  {
7      fstream file;
8      file.open("task.txt", ios::out);
9      if (!file)
10     {
11         cout << "Error in opening file";
12         return 1;
13     }
14     else
15     {
16         string line;
17         cout << "Enter a line of text: ";
18         while (getline(cin, line) && line != "")
19         {
20             cout << "Enter a line of text: ";
21
22             file << line << endl;
23         }
24     }
25     file.close();
26     file.open("task.txt", ios::in);
27     if (!file)
28     {
29         cout << "Error in opening file";
30         return 1;
31     }
```

```
32     else
33     {
34         string line;
35         int lineCount = 0;
36         while (getline(file, line))
37         {
38             lineCount++;
39         }
40         cout << "Number of lines in the file: " << lineCount << endl;
41     }
42     file.close();
43 }
```

```
Enter a line of text: Hello world
Enter a line of text: Hello World
Enter a line of text: Hello World
Enter a line of text: Hello World
Enter a line of text:
Number of lines in the file: 4
```

Ex-3

```
1  #include <iostream>
2  #include <fstream>
3
4  using namespace std;
5
6  class Student
7  {
8      double gpa;
9      const int id;
10     string name;
11     static int Scount;
12
13 public:
14     Student() : id(++Scount) {}
15     void getData()
16     {
17         cout << "Enter name: ";
18         cin >> name;
19         cout << "Enter GPA: ";
20         cin >> gpa;
21     }
22     void writeToFile(fstream &studentfile)
23     {
24         studentfile << "ID: " << id << " Name: " << name << " GPA: " << gpa << endl;
25     }
26 };
```

```
27 int Student::Scount = 0;
28 int main()
29 {
30     Student s[5];
31     fstream studentfile;
32     studentfile.open("students.txt", ios::out);
33     if (!studentfile)
34     {
35         cout << "Error in opening file";
36         return 1;
37     }
38     else
39     {
40         for (int i = 0; i < 5; i++)
41         {
42             s[i].getData();
43             s[i].writeToFile(studentfile);
44         }
45         cout << "Data written to file successfully." << endl;
46     }
47     studentfile.close();
48 }
```

```
Enter name: Yash
Enter GPA: 3.2
Enter name: Khunais
Enter GPA: 3.5
Enter name: Mahboor
Enter GPA: 3.9
Enter name: Qasim
Enter GPA: 3.2
Enter name: Ali
Enter GPA: 2.8
Data written to file successfully.
```

FILE:

```
students.txt
1 ID: 1 Name: Yash GPA: 3.2
2 ID: 2 Name: Khunais GPA: 3.5
3 ID: 3 Name: Mahboor GPA: 3.9
4 ID: 4 Name: Qasim GPA: 3.2
5 ID: 5 Name: Ali GPA: 2.8
6
```

TASK-3

```
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4  class Student
5  {
6      double gpa;
7      const int id;
8      string name;
9      static int Scount;
10
11 public:
12     Student() : id(++Scount) {}
13     void getData()
14     {
15         cout << "Enter name: ";
16         cin >> name;
17         cout << "Enter GPA: ";
18         cin >> gpa;
19     }
20     void writeToFile(fstream &studentfile)
21     {
22
23         studentfile << "ID: " << id << " Name: " << name << " GPA: " << gpa << endl;
24     }
25     void display(fstream &studentfile)
26     {
27
28         string line;
29         while (getline(studentfile, line))
30         {
31             cout << line << endl;
32         }
33     }
34 };

```

```
35 int Student::Scount = 0;
36 int main()
37 {
38     Student s[5];
39     fstream studentfile;
40     studentfile.open("students.txt", ios::out);
41     if (!studentfile)
42     {
43         cout << "Error in opening file";
44         return 1;
45     }
46     else
47     {
48         for (int i = 0; i < 5; i++)
49         {
50             s[i].getData();
51             s[i].writeToFile(studentfile);
52         }
53         cout << "Data written to file successfully." << endl;
54         studentfile.close();
55         studentfile.open("students.txt", ios::in);
56         for (int i = 0; i < 5; i++)
57         {
58             s[i].display(studentfile);
59         }
60         cout << "Data displayed from file successfully." << endl;
61         studentfile.close();
62     }
63 }

```

```
Enter name: Yash
Enter GPA: 3.2
Enter name: Khunais
Enter GPA: 3.5
Enter name: Mahboor
Enter GPA: 3.9
Enter name: Qasim
Enter GPA: 4
Enter name: Ali
Enter GPA: 3
Data written to file successfully.
ID: 1 Name: Yash GPA: 3.2
ID: 2 Name: Khunais GPA: 3.5
ID: 3 Name: Mahboor GPA: 3.9
ID: 4 Name: Qasim GPA: 4
ID: 5 Name: Ali GPA: 3
Data displayed from file successfully.
```

Ex-4

```
Ex-4.cpp > main()
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4  int main()
5  {
6      fstream file("demo.txt", ios::out);
7      try
8      {
9          if (!file)
10         {
11             throw "Error in opening file";
12         }
13         else
14         {
15             file << "File is Working properly" << endl;
16             cout << "Data written to file successfully." << endl;
17             file.close();
18         }
19     }
20     catch (const char *message)
21     {
22         cout << message << endl;
23     }
24 }
```

```
Data written to file successfully.
```

FILE:

```
demo.txt
1  File is Working properly
```

FINAL TASK

```
1  #include <iostream>
2  #include <fstream>
3  #include <string>
4
5  using namespace std;
6
7  class Student
8  {
9      double gpa;
10     const int id;
11     string name;
12     static int Scount;
13
14 public:
15     Student() : id(++Scount) {}
16     void getData()
17     {
18         cout << "Enter name: ";
19         getline(cin, name);
20         cout << "Enter GPA: ";
21         cin >> gpa;
22         cin.ignore();
23     }
24     void add(fstream &studentfile)
25     {
26         studentfile << "ID: " << id << " Name: " << name << " GPA: " << gpa << endl;
27     }
```

```
28     static void search(fstream &studentfile)
29     {
30         string line;
31         string searchName;
32         cout << "Enter Name to search: ";
33         getline(cin, searchName);
34         bool found = false;
35         while (getline(studentfile, line))
36         {
37             if (line.find(searchName) != string::npos)
38             {
39                 cout << "Record found: " << line << endl;
40                 found = true;
41             }
42         }
43         try
44         {
45             if (!found)
46             {
47                 throw "Record not found.";
48             }
49         }
50         catch (const char *message)
51         {
52             cout << message << endl;
53         }
54     }
```

```
55  static void view(fstream &studentfile)
56  {
57      if (Scount == 0)
58      {
59          cout << "No records to display.\n";
60          return;
61      }
62      string line;
63      while (getline(studentfile, line))
64      {
65          cout << line << endl;
66      }
67  }
68  static void initializeCount()
69  {
70      fstream studentfile("sfile.txt", ios::in);
71      if (studentfile.is_open())
72      {
73          string line;
74          while (getline(studentfile, line))
75          {
76              Scount++;
77          }
78          studentfile.close();
79      }
80  }
81  };
```

```
82  void display()
83  {
84      cout << "1. Add Student\n";
85      cout << "2. View All Students\n";
86      cout << "3. Search Student\n";
87      cout << "0. Exit\n";
88  }
89  int Student::Scount = 0;
90  int main()
91  {
92      Student::initializeCount();
93      fstream sfile;
94      int choice = -1;
95
96      while (choice != 0)
97      {
98          display();
99          cout << "Enter Choice: ";
100         cin >> choice;
101         cin.ignore();
```

```
102         switch (choice)
103         {
104             case 1:
105                 sfile.open("sfile.txt", ios::app);
106                 try
107                 {
108                     if (!sfile)
109                     {
110                         throw "Error in opening file";
111                         return 1;
112                     }
113                     else
114                     {
115                         Student s;
116                         s.getData();
117                         s.add(sfile);
118                         sfile.close();
119                     }
120                 }
121                 catch (const char *message)
122                 {
123                     cout << message << endl;
124                 }
125
126                 break;
```

```
127         case 2:
128             sfile.open("sfile.txt", ios::in);
129             try
130             {
131                 if (!sfile)
132                 {
133                     throw "Error in opening file";
134                     return 1;
135                 }
136                 else
137                 {
138                     Student::view(sfile);
139                     sfile.close();
140                 }
141             }
142             catch (const char *message)
143             {
144                 cout << message << endl;
145             }
146             break;
```

```
147         case 3:
148             sfile.open("sfile.txt", ios::in);
149             try
150             {
151                 if (!sfile)
152                 {
153                     throw "Error in opening file";
154                     return 1;
155                 }
156                 else
157                 {
158                     Student::search(sfile);
159                     sfile.close();
160                 }
161             }
162             catch (const char *message)
163             {
164                 cout << message << endl;
165             }
166
167             break;
168         case 0:
169             cout << "Exiting....\n";
170             break;
171         default:
172             cout << "Invalid Option!\n";
173     }
174 }
175 }
176
```

```
1. Add Student
2. View All Students
3. Search Student
0. Exit
Enter Choice: 1
Enter name: Yash
Enter GPA: 2.0
1. Add Student
2. View All Students
3. Search Student
0. Exit
Enter Choice: 1
Enter name: Khunais
Enter GPA: 3.9
1. Add Student
2. View All Students
3. Search Student
0. Exit
Enter Choice: 1
Enter name: Khunais Baig
Enter GPA: 3.4
1. Add Student
2. View All Students
3. Search Student
0. Exit
Enter Choice: 2
ID: 1 Name: Yash GPA: 2
ID: 2 Name: Khunais GPA: 3.9
ID: 3 Name: Khunais Baig GPA: 3.4
1. Add Student
2. View All Students
3. Search Student
0. Exit
```

```
Enter Choice: 3
Enter Name to search: Khunais
Record found: ID: 2 Name: Khunais GPA: 3.9
Record found: ID: 3 Name: Khunais Baig GPA: 3.4
1. Add Student
2. View All Students
3. Search Student
0. Exit
Enter Choice: 0
Exiting....
```